

Abstract

KLH SmartLib is a web-based library management system designed for KLH University. It digitizes book and copy management, issue/return circulation, reservations, fines, users, notifications and reports. The system provides separate role-based workflows for students, librarians and administrators, with the student portal remaining within the student workflow. The proposed stack uses React, TypeScript, Vite and Tailwind CSS on the frontend, Flask with SQLAlchemy on the backend, and MySQL for persistent storage.

Tools & Technologies

Front-End: React 19, TypeScript, Vite, Tailwind CSS
Back-End: Python Flask + SQLAlchemy
Database: MySQL
Authentication: JWT-based role authentication
Library Operations: Books, copies, loans, returns, reservations and fines
Development: VS Code, Git, GitHub
Additional: Barcode/RFID-ready circulation, analytics and reporting

Problem Statement & Objectives

Problem: Traditional/manual library workflows can make circulation, inventory tracking, reservations, fines and reporting difficult to manage consistently.

Objective 1: Provide secure, role-based student, librarian and admin workflows.
Objective 2: Manage books, individual copies, categories and physical locations.
Objective 3: Automate issue/return, reservations, fine calculation and status tracking.
Objective 4: Maintain centralized records, notifications, reports and audit information.
Scope: Authentication, catalog search, digital library cards, circulation, reservations, fines, notifications, analytics and administration.

System Architecture

USER
↓
React + TypeScript + Vite + Tailwind Web Interface
↓
Flask REST Backend + SQLAlchemy
↓
JWT Authentication + Role-Based Access Control
↓
MySQL Database
↓
Users • Books • Copies • Loans • Reservations • Fines
↓
Notifications • Reports • Analytics • Audit Logs

Existing vs Proposed System

Feature	Traditional / Existing	KLH SmartLib
Book tracking	Manual registers / basic records	Book + copy-level digital tracking
Circulation	Manual issue and return entries	Centralized loan and return workflow
Reservations	Manual queue handling	Reservation queue with status tracking
Fines	Manual calculation	Automatic fine calculation and records
User access	Less centralized role control	Separate Student / Librarian / Admin workflows
Reporting	Manual preparation	Digital reports, analytics and audit records

ER Diagram & Database Design

USERS	user_id (PK), name, email, role
BOOKS	book_id (PK), title, ISBN, category_id, author, publisher
BOOK_COPIES	copy_id (PK), book_id (FK), barcode, location_id, status
LOANS	loan_id (PK), user_id (FK), copy_id (FK), issue_date, due_date, return_date, fine_id
RESERVATIONS	reservation_id (PK), user_id (FK), book_id (FK), status, reserved_at
FINES	fine_id (PK), user_id (FK), loan_id (FK), amount, status
CATEGORIES / LOCATIONS	category_id / location_id; DDC, floor, section, rack, shelf

Relationships: User→Loans/Reservations/Fines; Book→Copies/Reservations; Copy→Loans; Loan→Fine.
Location supports floor/section/rack/shelf organization; role-based access separates student and staff workflows.

Feasibility Study

Technical: React/TypeScript frontend with Flask, SQLAlchemy and MySQL provides a practical full-stack architecture for a university library system.
Economic: Uses widely available open-source development tools and a relational database; development can be performed on a standard system.
Operational: Separate student, librarian and admin workflows support clear responsibilities and easier maintenance.
Scalability: The relational design supports books, copies, circulation, reservations, fines, reporting and future barcode/RFID integrations.

Core Modules / Implementation

- Student Portal — catalog search, availability, digital library card, loans and reservations
- Librarian Portal — circulation, copy tracking, returns, reservations and fine handling
- Admin Portal — users, categories, locations, policies, reports and audit controls
- Catalog — books, authors, ISBN, categories and DDC-based classification
- Circulation — issue/return, due dates, automatic fine records and status updates
- Analytics — circulation trends, inventory views and operational reports

Results & Testing

The proposed system brings core library operations into a centralized database-backed workflow.

Testing areas:

- Role-based authentication
- Book and copy management
- Issue / return workflows
- Reservations and fine records
- Student, librarian and admin access separation
- Database CRUD and reporting operations

Expected outcome: consistent records and easier library administration.

Conclusion

KLH SmartLib provides a structured digital library platform connecting users, catalog records, physical copies and circulation data through a relational database.

The design supports secure role-based workflows, centralized library records, reservations, fines, notifications, analytics and future barcode/RFID integration.

Future Scope

1. Integrate barcode/RFID scanning for faster circulation.
2. Add email/SMS/push notifications for due dates and reservations.
3. Add cloud deployment and automated backup.
4. Add richer analytics and recommendation features.
5. Integrate digital resources and e-library access.
6. Expand audit, security and administrative controls.